Cognitive Computing

Introduction to Natural language processing (NLP)

Role of NLP in Cognitive System



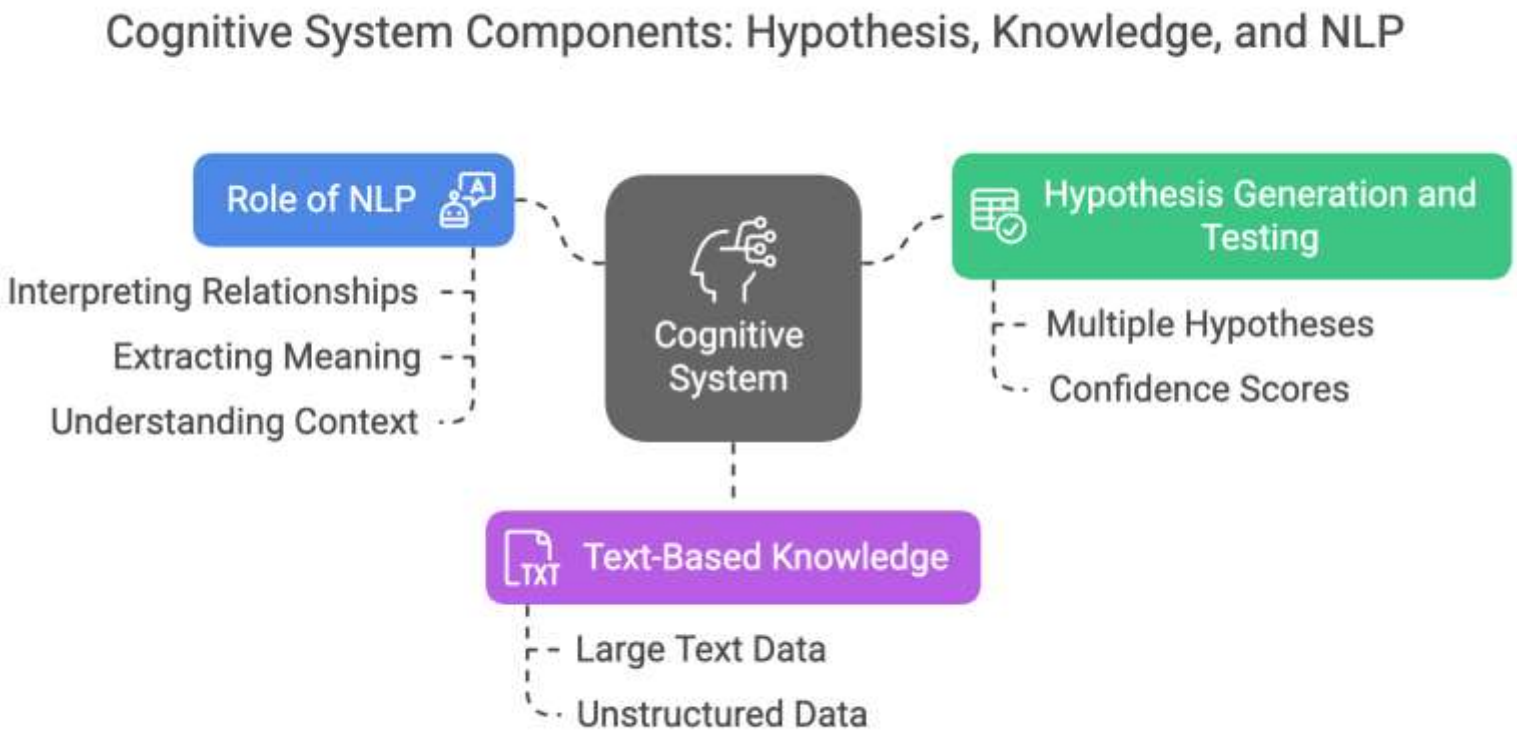
- **Cognitive System**
  - *Mimic human-like reasoning and decision making*
  - *Generate and Score Hypothesis*
  - *Alternative answers – each with a degree of confidence*

Cognitive system vs other data-driven techniques

- **Capability to manage, understand, and analyze unstructured data in context with the questions being asked. (rely on large amount of text data)**
- **Unstructured information must be parsed and tagged to find the elements of meaning**
- **Understand meaning and intent behind text**
- **Extensive NLP for human like responses**
- **E.g.: IBM Watson, ChatGPT**

- **Works with Structured Data (like tables) and relies on schemas to add context and meaning to data**
- **Identifies patterns but lacks deeper understanding**
- **Limited NLP**
- **SQL-based systems, basic ML models**

Role of NLP


Cognitive System Components: Hypothesis, Knowledge, and NLP
Role of NLP
Interpreting Relationships
Extracting Meaning
Understanding Context
Cognitive System
Hypothesis Generation and Testing
Multiple Hypotheses
Confidence Scores
Text-Based Knowledge
Large Text Data
Unstructured Data

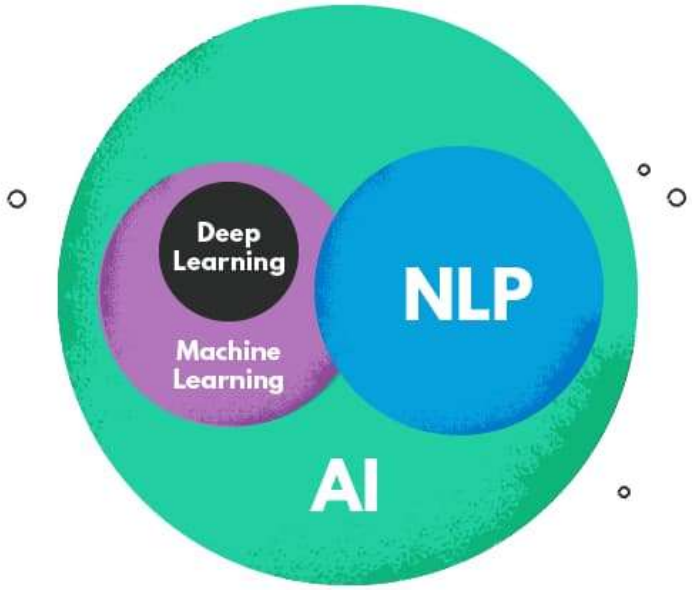
Role of NLP in Cognitive System (contd..)

- Tools for this process of identifying the meaning of the individual words include categorization, ontologies, tagging, catalogs, dictionaries, and language models (NLP).
- This is where **Natural Language Processing (NLP) techniques** come in.

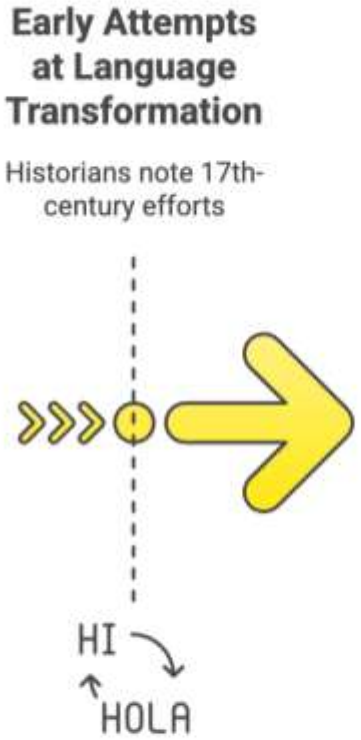
Remember - Not all unstructured data is text.
There is a requirement in some cognitive computing systems to support images, video, speech, and sensor data, depending on how the data will be used.

Natural Language Processing (NLP)

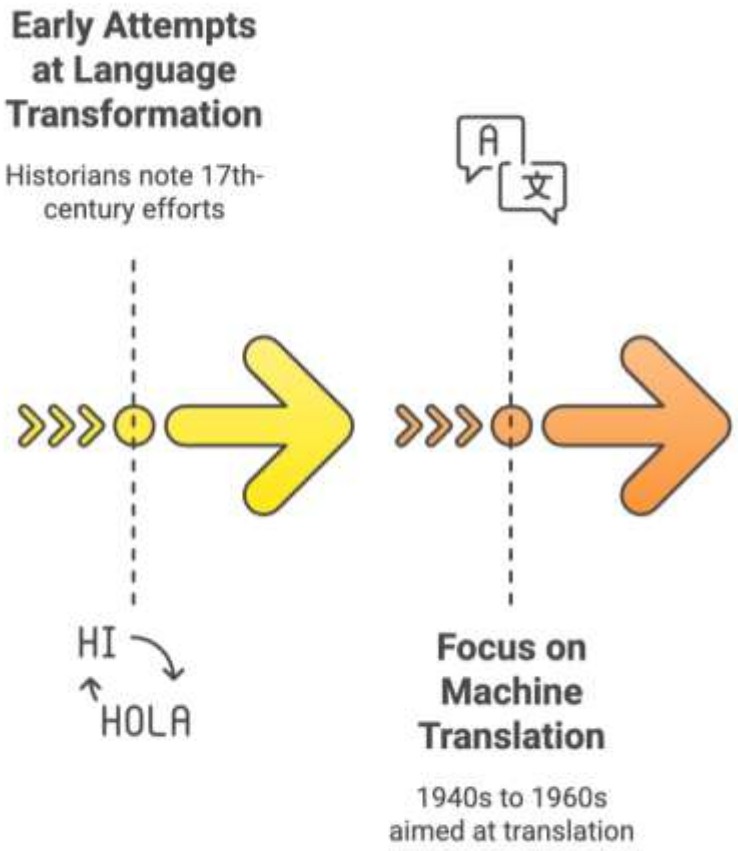
- A field of artificial intelligence (AI) that enables computers to understand, interpret, and generate human language.
- It combines linguistics, machine learning, and computational techniques to process and analyze large amounts of text and speech (unstructured) data.
- NLP – Set of techniques that extract meaning from text



THE HISTORY OF NLP



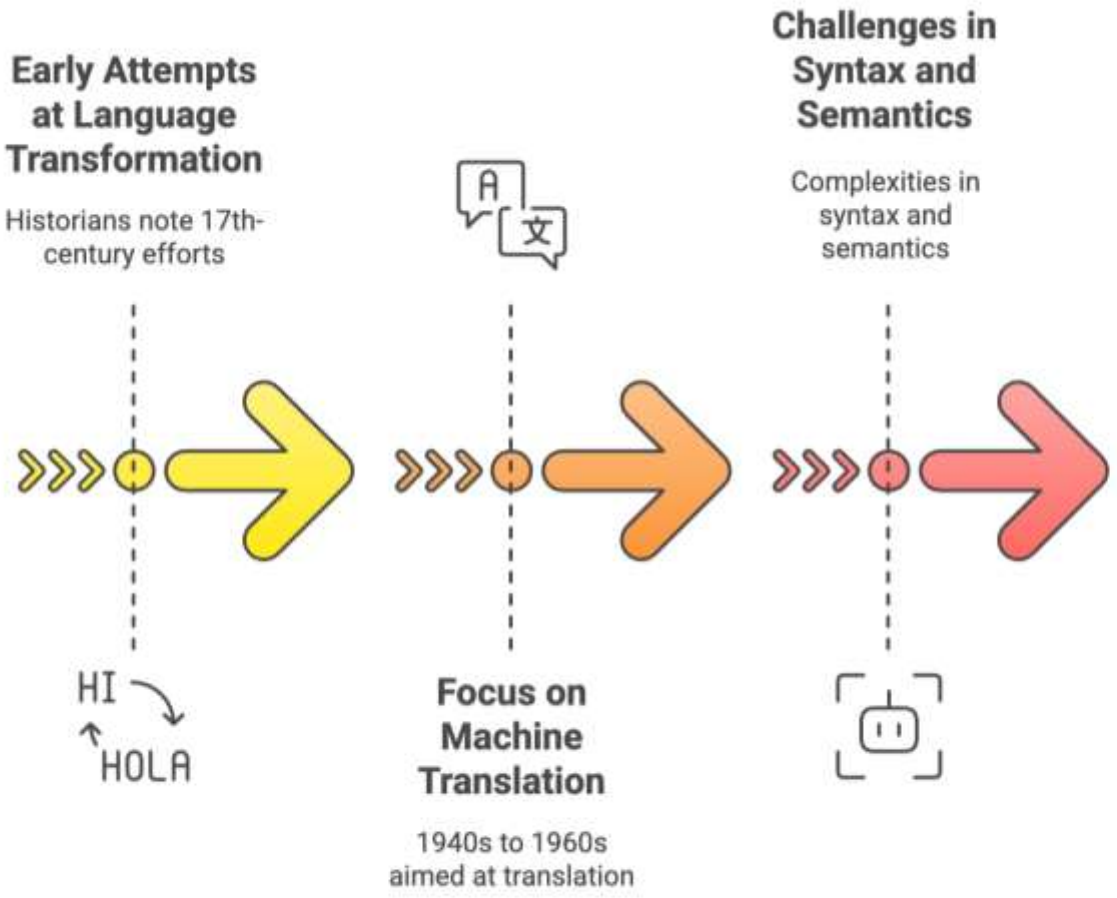
THE HISTORY OF NLP



- Early approaches relied on word-to-word dictionary translations, where each word was looked up in a bilingual dictionary and replaced with its counterpart.

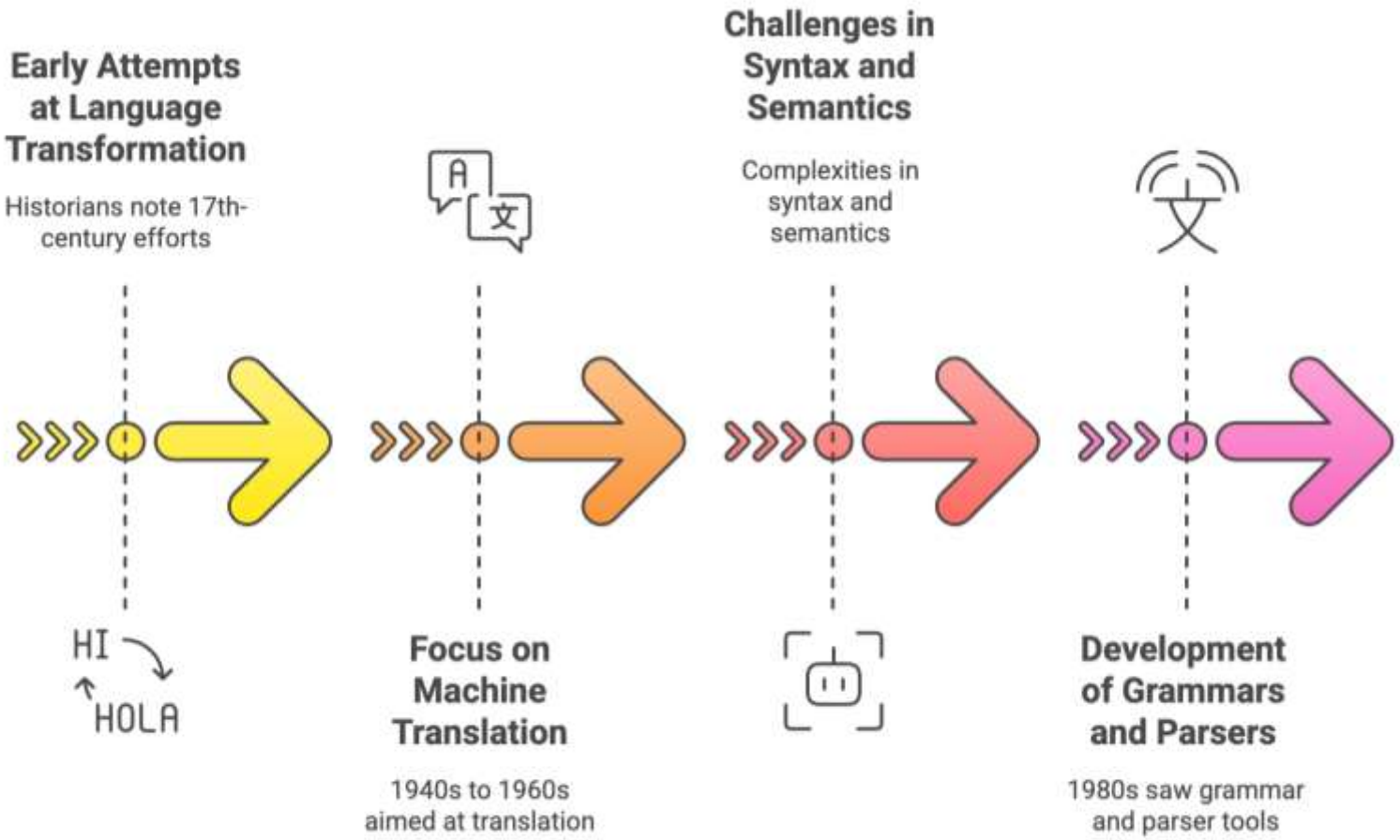
- **Challenges:**
  - This method was slow and inefficient.
  - It failed to capture sentence structure, syntax, and meaning (e.g., idioms and context-dependent words).

THE HISTORY OF NLP



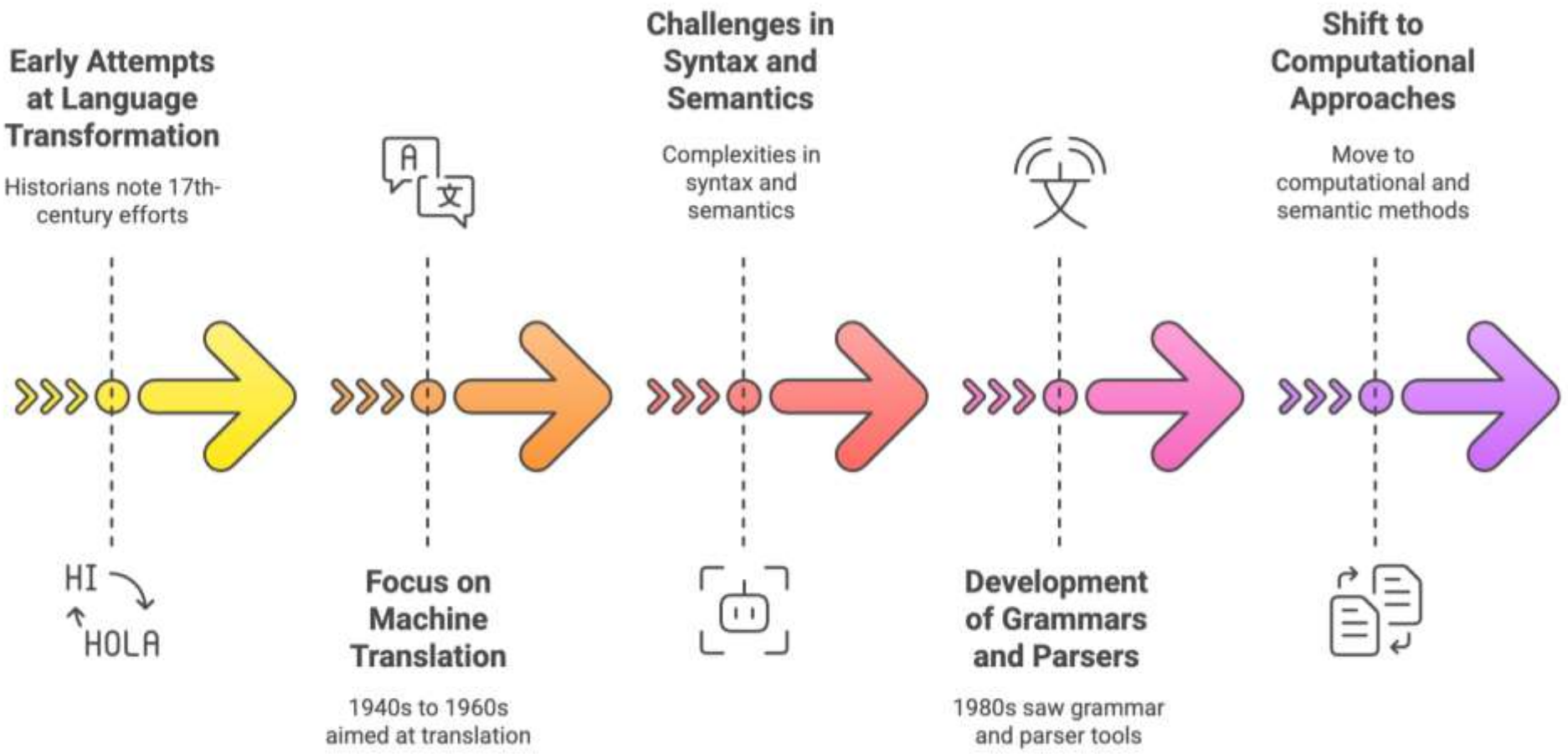
- ***Syntactic processing (understanding sentence structure).***
- ***Semantic processing (understanding meaning and context).***
- ***Direct word-for-word translation often led to incorrect or unnatural sentences.***

THE HISTORY OF NLP



- ***Parsers & Grammars – Systems began analyzing not just words but also sentence structure and rules.***
- ***Word Sense Disambiguation (WSD) – Understanding words with multiple meanings (e.g., "bank" as a financial institution vs. "bank" of a river).***

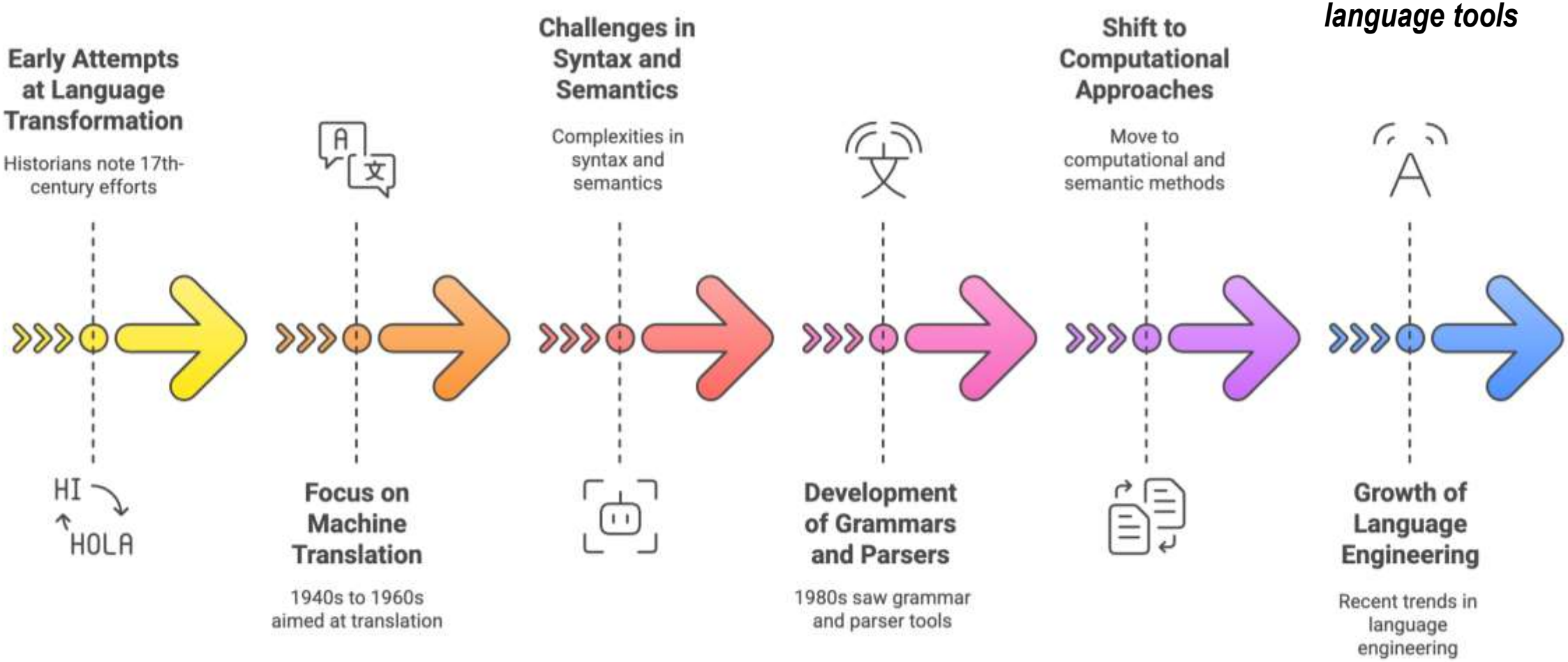
THE HISTORY OF NLP



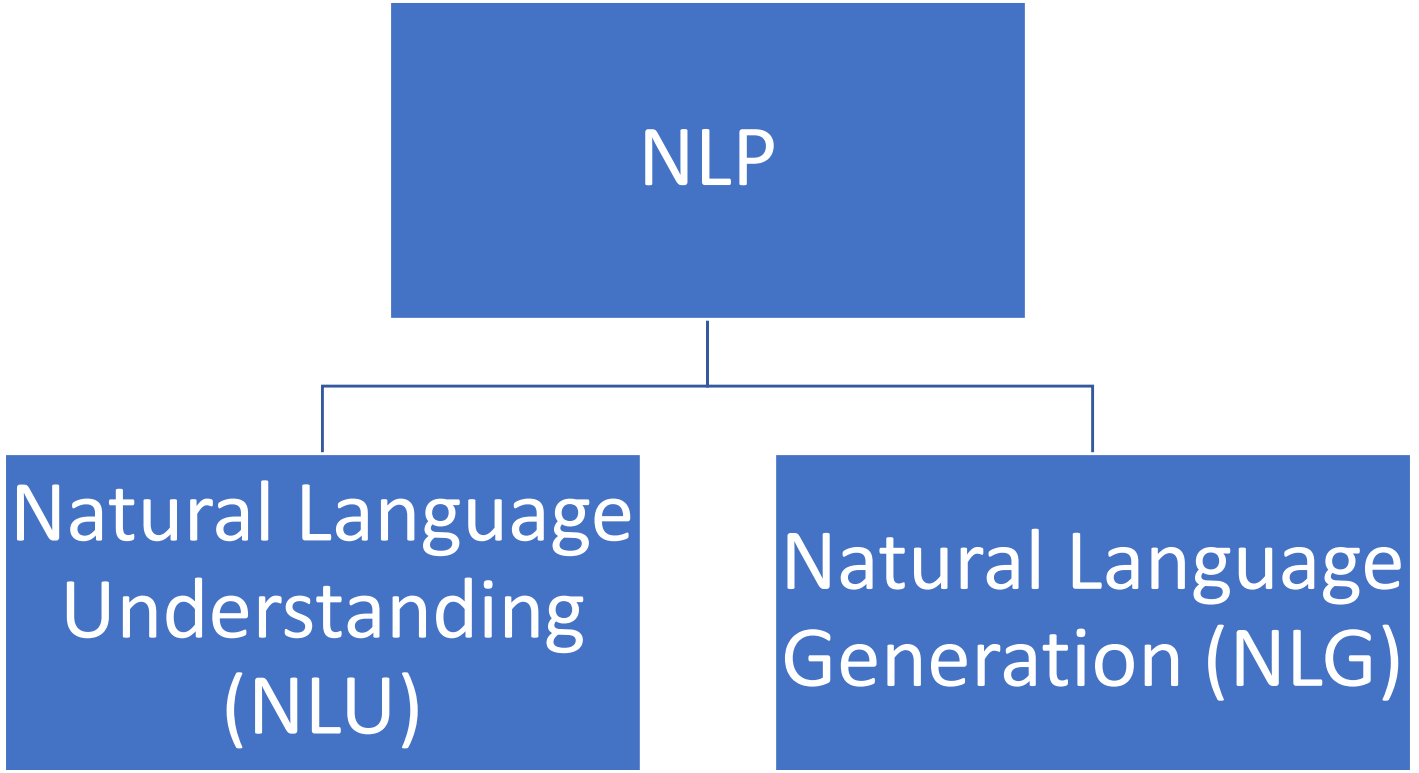
- ***Probabilistic Models & Statistical Methods – Instead of relying on fixed rules, systems used probabilities to improve accuracy.***
- ***Impact: Marked the transition from a mechanical approach (rule-based translation) to a computational and semantic approach (understanding meaning and context).***

THE HISTORY OF NLP

About applying NLP techniques to create functional, real-world language tools



Broad Categories of NLP



NLU refers to a process by which an inanimate object with computing power is able to comprehend spoken language

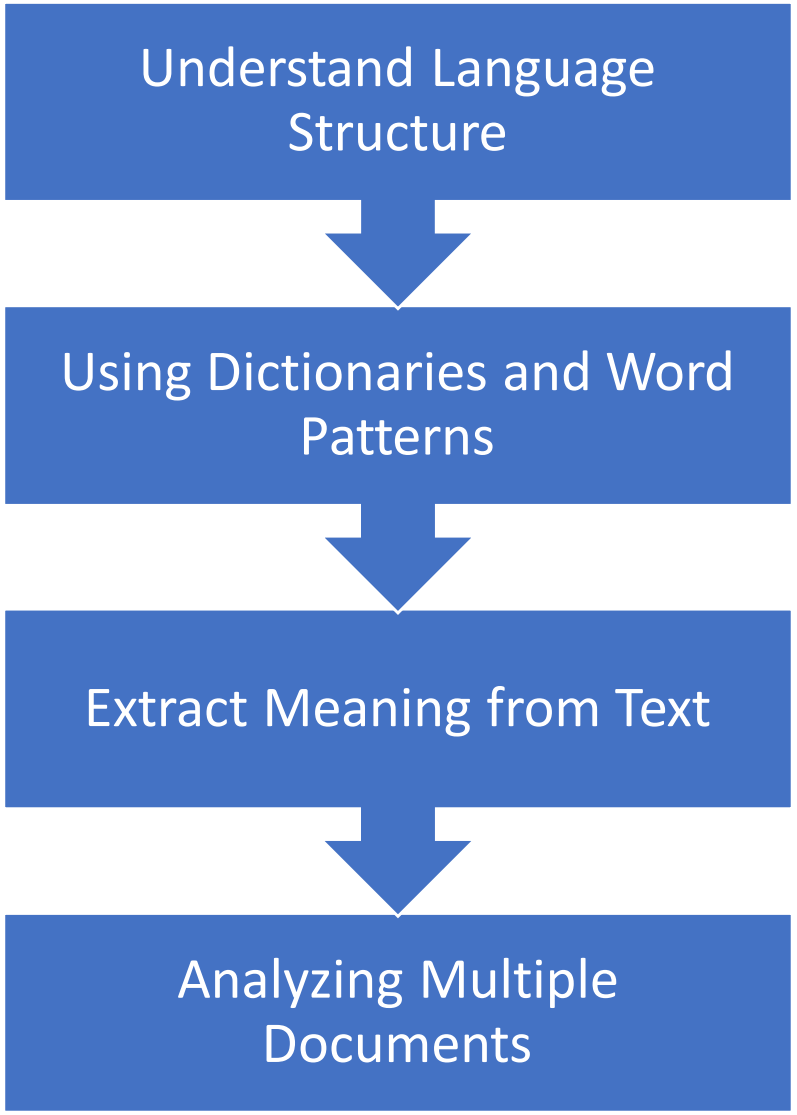
NLG refers to a process by which an inanimate object with computing power is able to manifest its thoughts in a language that humans are able to understand

Text Analytics vs NLP

- Text Analytics - The art of extracting useful insights and answering questions from any given text data
- NLP – Not just restricted to text data. Voice (speech) recognition and analysis also come under domain of NLP.



How does NLP work? General Framework



NLP systems analyze language by recognizing patterns, structures, and meanings in words and sentences.

- Example: "The cat is sleeping on the mat."
- NLP identifies "cat" as a noun, "sleeping" as a verb, and "on the mat" as location information.

Computers use dictionaries (word meanings) and patterns of how words appear together to understand context.

- Example: "Apple is a fruit." → Here, Apple refers to food.
- "Apple released a new iPhone." → Here, Apple is a company.
- NLP can recognize the difference using context!

NLP can find important information like names, locations, events, and actions from a document. Example: If you give a computer a news article about "Elon Musk launching a SpaceX rocket from Florida", NLP can extract:

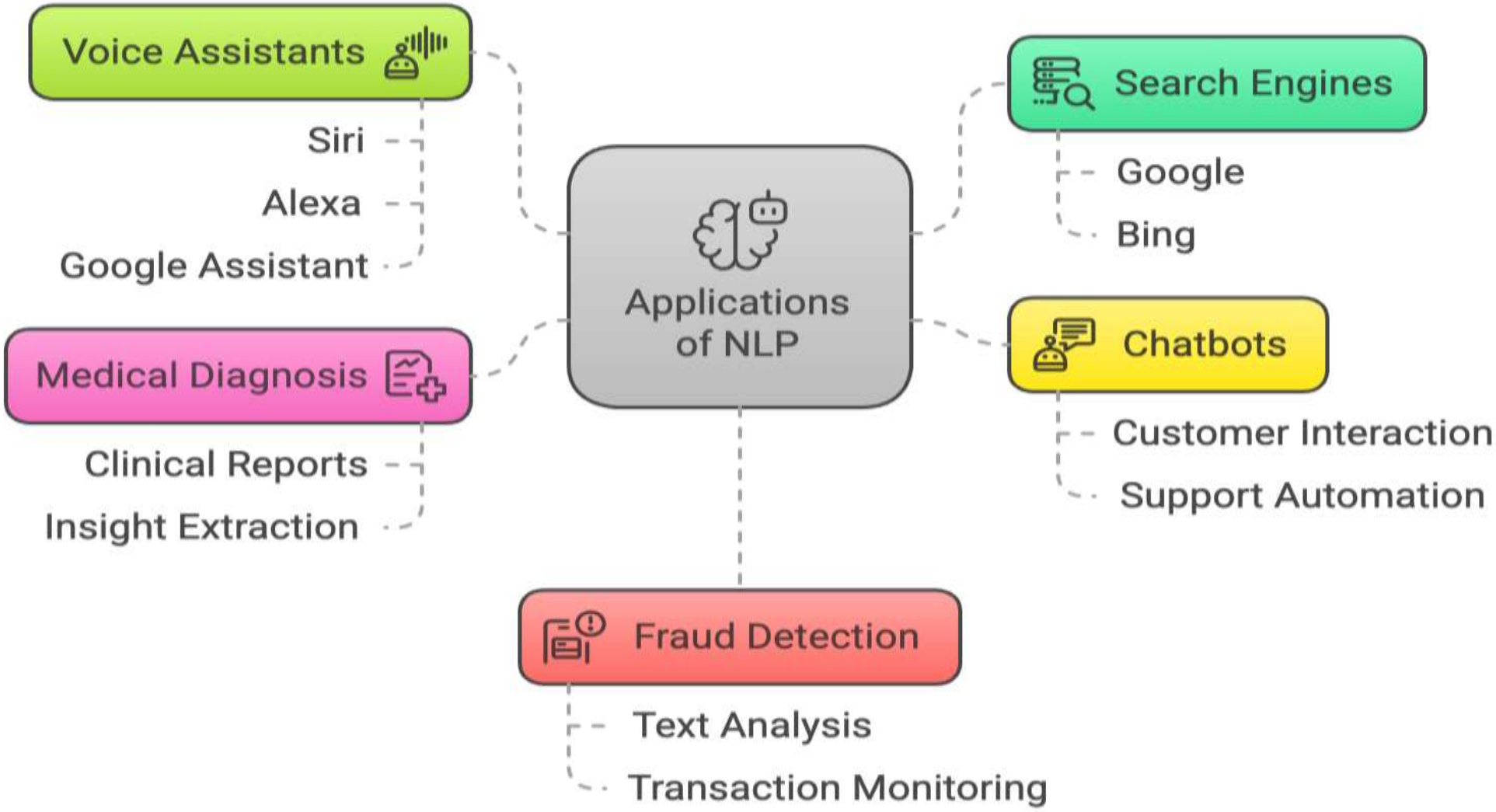
- Person: Elon Musk; Company: SpaceX; Action: Launching; Location: Florida

- NLP is not limited to one sentence—it can compare and analyze multiple documents!
- Example: If two news articles mention "the same accident but use different wording", NLP can still identify it's the same event.

Example

London is the capital and most populous city of England and the United Kingdom. Standing on the River Thames in the south east of the island of Great Britain, London has been a major settlement for two millennia. It was founded by the Romans, who named it Londinium. London's ancient core, the City of London, largely retains its 1.12-square-mile (2.9 km2) medieval boundaries.

Applications of NLP



Understanding Linguistics

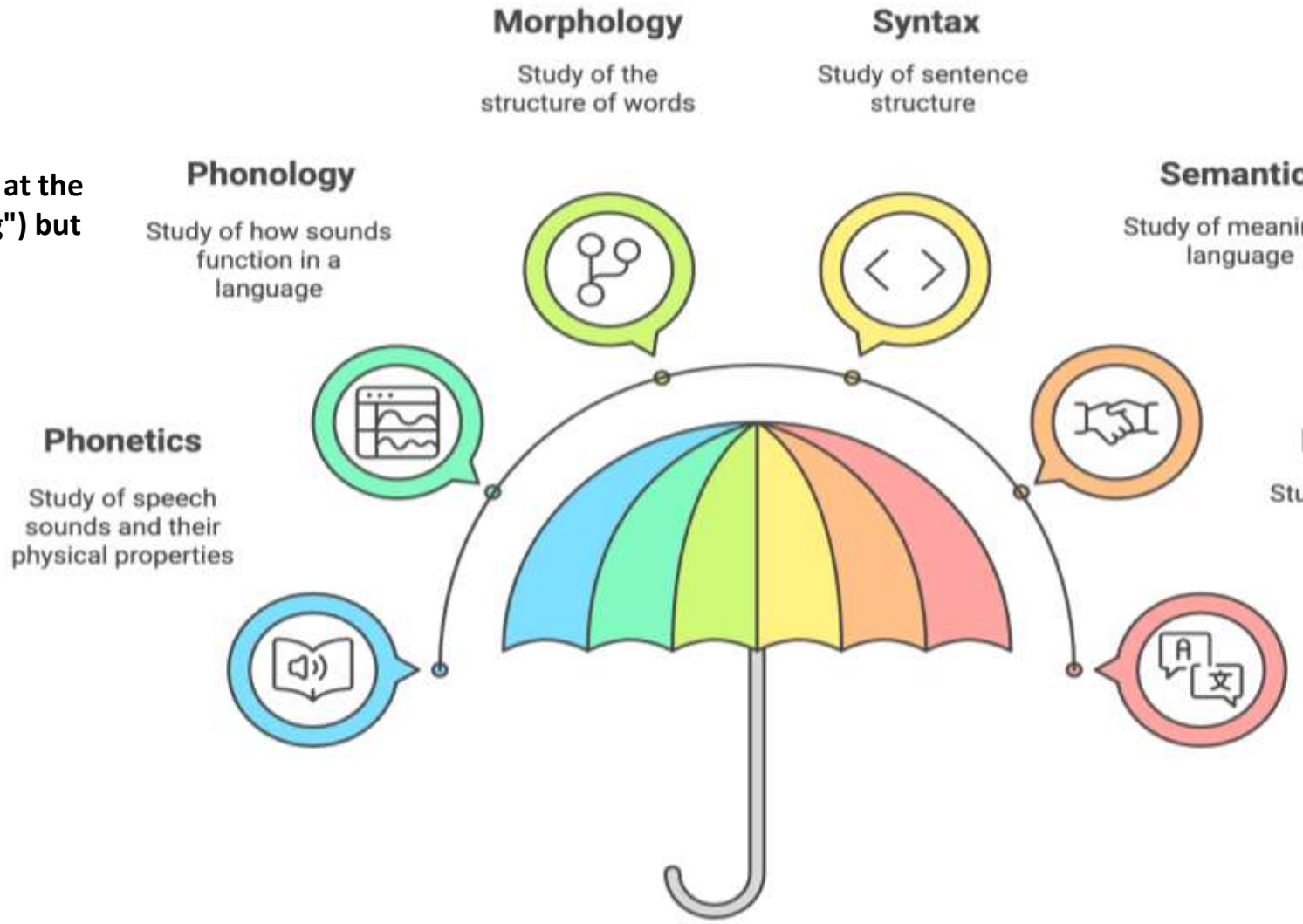
- NLP = Linguistics + ML + Computational Techniques
- Linguistics – The Study of Language
  - Linguistics is the scientific study of language and its structure, including how it is formed, used, and understood.
  - It explores both the rules that govern languages and how humans acquire and process them.

Branches of Linguistics/NLP Terminology

Replaying" → "re-" (prefix) + "play" (root) + "-ing" (suffix).

In English, "ng" appears at the end of words (e.g., "sing") but never at the beginning.

difference between pronouncing "p" (with a puff of air) and "b" (without a puff of air).
pitch, loudness, frequency



"The cat chased the mouse" vs. "The mouse chased the cat" (same words, different meanings).

"Can you open the window?" is a request, not just a question about ability.

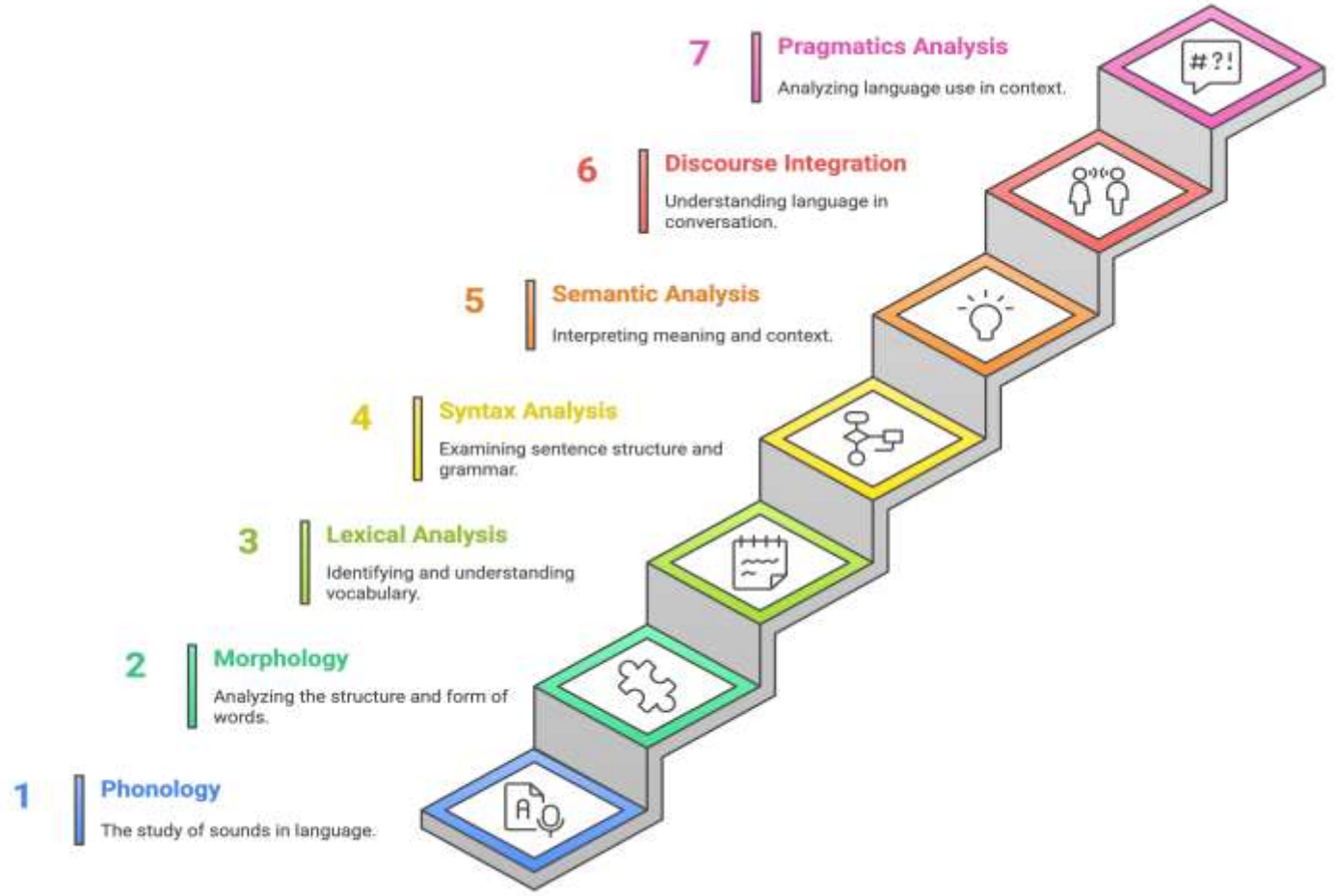
Role of Linguistics in NLP

Role in NLP :

- Linguistic analysis breaks down the text to provide meaning.
- Linguistics provide rules and structures in NLP that help AI systems understand and generate human language.
  - For example:
    - Syntax & Parsing help computers understand sentence structure.
    - Semantics improves meaning extraction in chatbots.
    - Phonetics & Speech Processing aid in voice recognition technologies.

Main Steps/Phases/Levels of NLP

1. Phonology
2. Morphology
3. Lexical Analysis
4. Syntax Analysis
5. Semantic Analysis
6. Discourse Integration
7. Pragmatics Analysis



Phonology in NLP

- Phonology is the study of the physical sounds of a language and how those sounds are uttered in a particular language.
- This area is important for speech recognition and speech synthesis but is not important for interpreting written text.
- However, to understand, for instance, the soundtrack of a video, or the recording of a call center call, not only is the pronunciation of the words important (regional accents such as British English or Southern United States), but the intonation patterns.
- A person who is angry may use the same words as a person who is confused; however, differences in intonation will convey differences in emotion.

Phonology in NLP

- **Phonemes** – The smallest unit of sound that can change meaning.
    - Example: In English, **/p/** and **/b/** are phonemes because they distinguish words like *pat* and *bat*.

- Why is Phonology important?
    - Helps in understanding **speech patterns** and **accents**.
    - Essential for **language learning** and **speech therapy**.
    - Useful in **speech recognition** and **text-to-speech** systems (e.g., Siri, Google Assistant).

Morphology

- **Morphology refers to the structure of a word.**
- **Morphology breaks words into morphemes (smallest units of meaning).**
- **Types of Morphemes:**
  - ***Root Morpheme:*** *The base word (e.g., "happy" in "unhappiness").*
  - ***Affixes:*** *Prefixes and suffixes that modify meaning (e.g., "un-" (prefix) and "-ness" (suffix) in "unhappiness").*

Step 1. Lexical Analysis/ Text Preprocessing

- In any analysis of incoming text, the first process is to identify which language the text is written in and then to break the text apart into separate sentences.
- Later, separate the sentences into words (Tokenization).
- Step 1.1 Tokenization: Process of splitting a sentence into its constituents i.e. separate words or tokens.
- Eg. "I am reading a book." ----------> "I", "am", "reading", "a", "book", "."

These tokens are called Unigrams (one token at a time)

n-gram refers to a sequence of n items from a given text.

- Unigrams – One token at a time
- Bigrams- Extract two tokens at a time
- Trigrams – Extract three tokens at a time

Eg: I would love to visit the United States
Unigram can give wrong results here as United States must be treated as single entity.

Lexical Analysis – Issues with Tokenization

1. Ambiguities that arise due to whitespaces, hyphens, contractions, etc.
   - Many languages like Chinese and Japanese do not separate words with spaces.

     Chinese Text:

     我喜欢看电影(I like watching movies)

     Possible tokenizations:

     - ["我" (I), "喜欢" (like), "看" (watch), "电影" (movie)] -> Right
     - ["我" (I), "喜欢看" (like watching), "电影" (movie)] -> Wrong
   - Multi-word expressions(MWE)

     English Phrases:

     "New York" may be split into ["New", "York"] instead of ["New York"].

     Idioms:

     "Kick the bucket" (meaning "to die") should not be split into ["Kick", "the", "bucket"].

Lexical Analysis – Issues with Tokenization

1. Ambiguities that arise due to whitespaces, hyphens, contractions, etc.
   - Contractions.
     - **English Contractions:** The appropriate choice depends on the tokenization rules and the NLP application.
       - *"don't"* → Should it be ["do", "n’t”] or ["don’t"]?
       - *"I'm"* → Should it be ["I", "'m"] or ["I'm"]?
   - Punctuation Issues
     - **Sentence:** *"Dr. Smith is a Ph.D. holder."*
       Incorrect: ["Dr", ".", "Smith", "is", "a", "Ph", ".", "D", ".", "holder", "."] ✗
       Correct: ["Dr.", "Smith", "is", "a", "Ph.D.", "holder", "."] ✓
   - Numbers and Dates

     Date: *"12/05/2024"*
     ["12", "/", "05", "/", "2024"] ✗
     ["12/05/2024"] ✓
     Number: *"$10.5 million"*
     ["$", "10", ".", "5", "million"] ✗
     ["$10.5", "million"] ✓

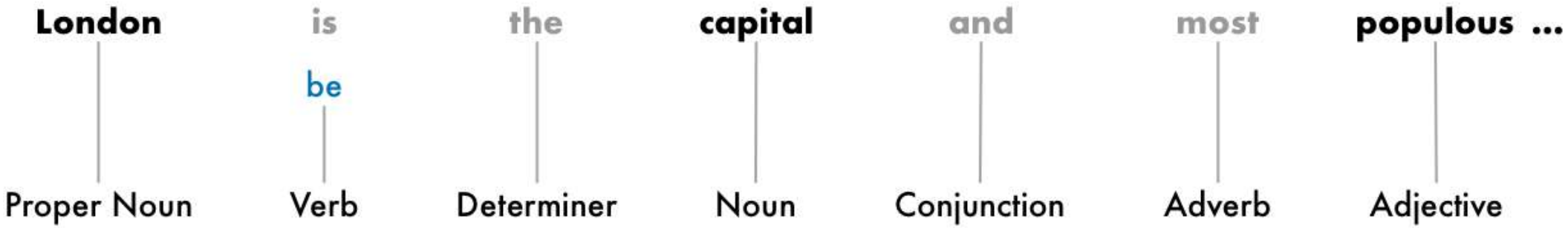
- For example, if you are building a rock band search engine, you want to make sure you don't ignore the word "The". Because not only does the word "The" appear in a lot of band names, there's a famous 1980's rock band called *The The*!



Step 1. Lexical Analysis/ Text Preprocessing

Step 1.2: StopWord Removal (Later -> Zipf's Law)

- Stop words - Common words that are just used to support construction of sentences
  - Do not impact meaning of sentence and can add noise
  - Remove the most commonly occurring words that don't add much information to the text. For example, "the," "a," "an," and so on.
  - There's no standard list of stop words that is appropriate for all applications



Stop words grayed out

Another example: I am happy vs I am not happy

Statistical Nature of Languages

- We now know : *how text can be processed linguistically (like lexical-> stemming/lemmatization -> syntactic -> semantic ->……)*

Can we understand text statistically?

- **Sentence 1:** The cat is sitting on the mat.
- **Sentence 2:** The cat is sleeping on the mat.

Words repeated: the, is, on, mat, cat

Some words change: sleeping, sitting

*Words occur with **different frequencies.***
NLP makes extensive use of this statistical property.

Word Frequency

- Number of times a particular word occurs in a text or corpus
- Example Corpus:
  - *D1: AI is useful.*
  - *D2: AI is powerful.*
  - *D3: AI helps humans.*



Word	Frequency
AI	3
is	2
useful	1
powerful	1
helps	1
humans	1

- *Example 2:* Suppose a document with 10,000 words
  - The word: **"the"** occurs 800 times.
  - The word: **"quantum"** occurs 15 times.

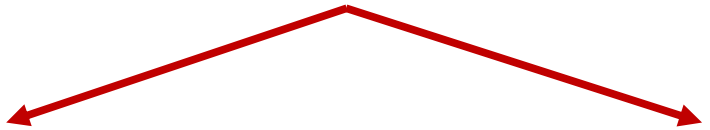
Does a high frequency automatically mean that a word is important?

NO!

Token vs Type

- **Token** - an individual occurrence of a word in the text
- **Type -** a unique word

cat dog cat bird dog cat



Tokens – 6

Tokens: {cat dog cat bird dog cat}

Gives corpus length

Type – 3

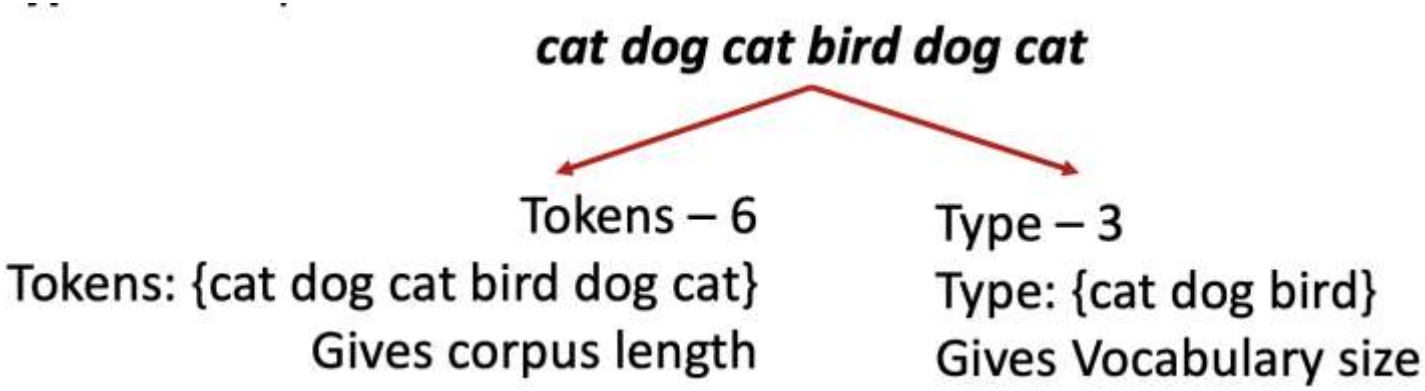
Type: {cat dog bird}

Gives Vocabulary size

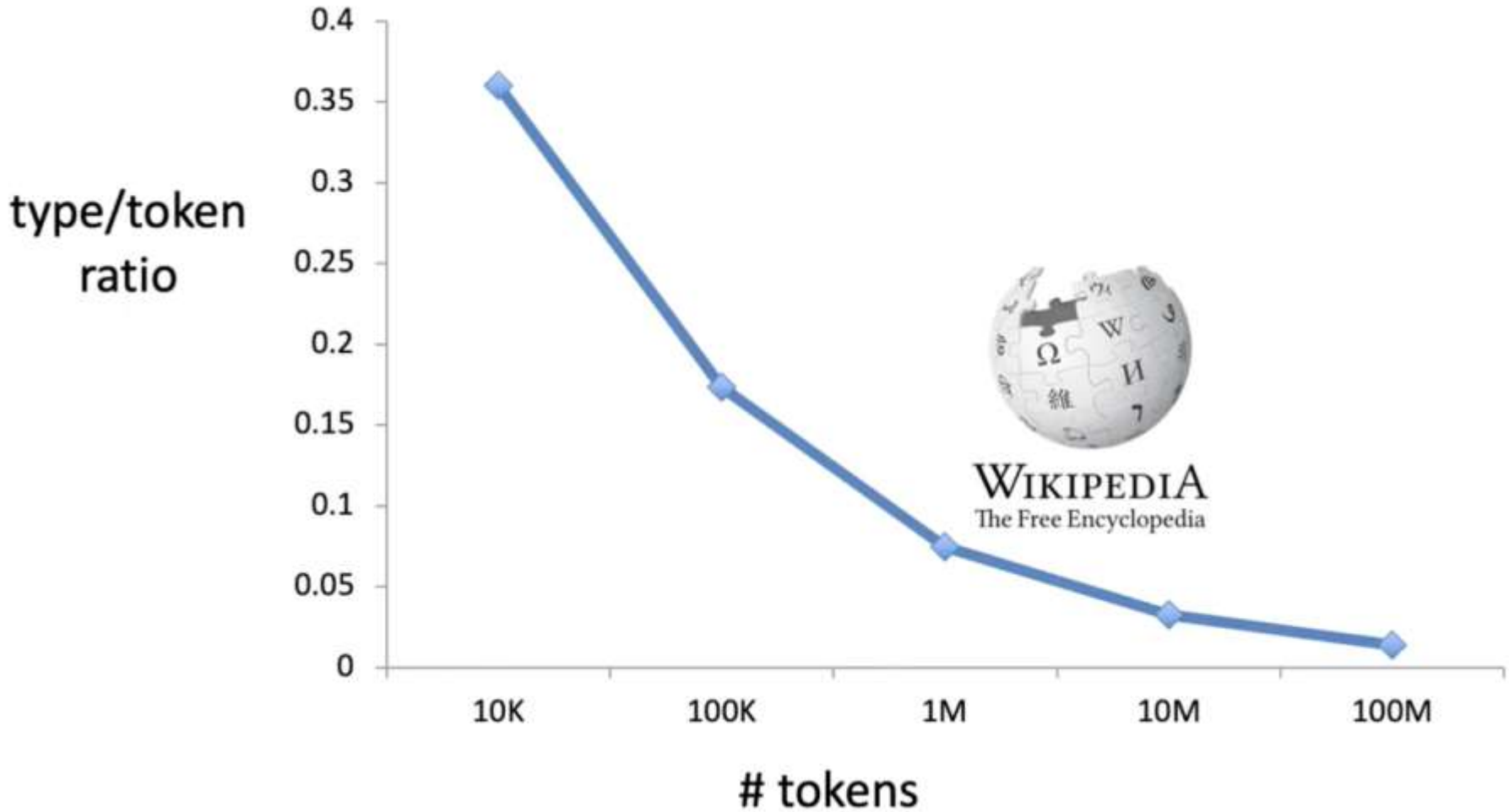
Why It Matters in NLP

- **Vocabulary Size (Types):** Tells the model how many distinct words it needs to learn.
- **Corpus Length (Tokens):** Measures the total size or volume of the text processed.
- **Type-Token Ratio (TTR):** Dividing types by tokens gives you the TTR, which helps measure text vocabulary richness and complexity.

- Here, TTR = 3/6 = 0.5



How does the type/token ratio change when adding more data?


type/token ratio
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
10K
100K
1M
10M
100M
tokens
WIKIPEDIA
The Free Encyclopedia



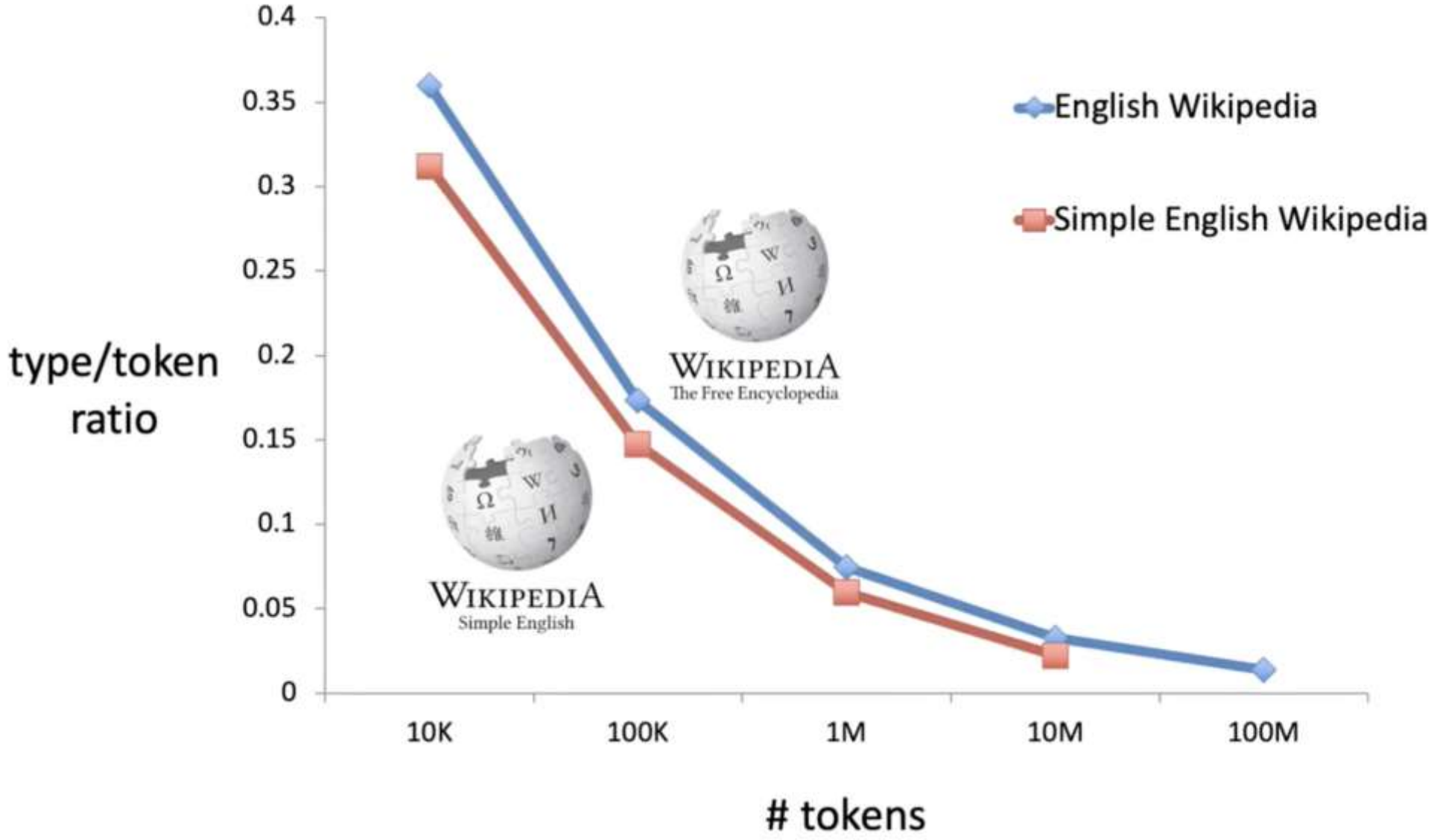
WIKIPEDIA
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VS.



WIKIPEDIA
Simple English

Which has a higher type/token ratio?



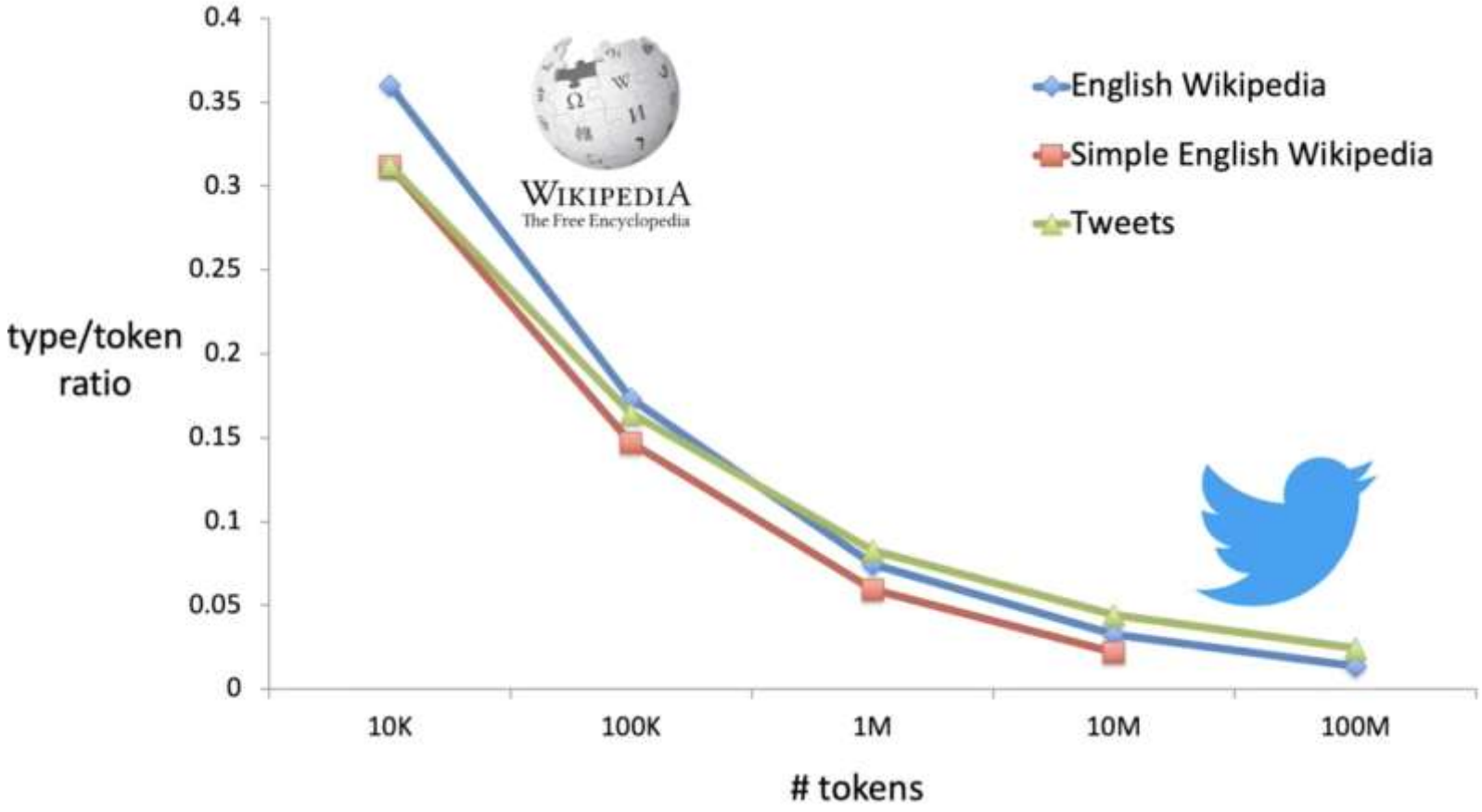
Higher TTR is a measure of complexity of the corpus and indicates greater lexical variety within the analyzed text.



VS.



For small dataset, twitter has smaller TTR… but its TTR increases as more data is added..



WHY?

- Because twitter shows enormous amounts of variations in word forms

224571 really	38 really2	12 reaaaaaally	7 reeeealy	5 rrly	3 realiy
1189 rly	37 reaaaaally	12 rreally	7 reeeeeeeeeally	5 rrrreally	3 realllllllllllllly
1119 realy	35 reallyyyyy	11 reaallyy	7 relaly	4 reaaaaly	3 reallllllly
731 rlly	31 reely	11 reallllyyy	6 r-e-a-l-l-y	4 reaaalllly	3 reallllllyyyy
590 reallly	30 realllyyy	11 reeeeallly	6 r-really	4 reaaalllyy	3 reallllllyyyyyyy
234 realllly	27 reaaly	11 reeeeallly	6 reaaaaaallly	4 reaalllly	3 realllllyyyyy
216 reallyy	27 realllyy	10 reaaaly	6 reallllllllly	4 reaalllyyy	3 realluy
156 relly	26 realllyyyy	10 reallyreallyreally	6 realllyyyyy	4 reallllllyyyy	3 really)
146 reallllly	25 reallllllllly	9 r)eally	6 realyl	4 reallllllyyyy	3 reallyl
132 rily	22 reaaallly	9 really-really	6 reeeaaaally	4 reeaaaally	3 reallyyyyyyyyy
104 reallyyy	21 really-	9 reallys	6 reeeaaallly	4 reeealy	3 reeaaallly
89 reallllllly	19 reeaally	9 reeeeeeeally	6 reeeaaalllyyy	4 reeeeeeeeeeally	3 reeaalllly
89 reeeally	18 reallllyyy	8 realky	5 reaaaaallly	4 rllly	3 reeaalllyyy
84 reaaally	16 reaaaallly	8 reallyyyyyy	5 reaaaalllly	3 r34lly	3 reeaaly
82 reaally	15 reaallly	8 reallyyyyyyy	5 reaalllyy	3 r]eally	3 reeallly
72 reeeeally	15 reallllllllly	8 reeeaally	5 realllllllllllly	3 reaaaaaaaally	3 reealy
65 reaaaally	15 realllyy	7 r3ally	5 realllllllllly	3 reaaaaaly	3 reeeaaalllyyy
57 reallyyyy	15 reallyreally	7 raelly	5 reeallyyy	3 reaaaalllly	3 reeeaallly
53 rilly	15 realyy	7 reaaaaaaally	5 reeeeaaallly	3 reaaaallyy	3 reeeaaaaaally
50 reallllllly	14 realllyyyy	7 reallllllllllly	5 reeeeaaally	3 reaaallyy	3 reeeeallllly
48 reeeeeally	14 reeeeeeally	7 realllllllyyy	5 reeeeeeeeeally	3 reaalllllly	3 reeeeealllly
41 reeally	13 reeeaaally	7 reeeeaaally			

How many words are there?

- Size of vocabulary continues to grow as you collect more data
- You will never find all the words
- How words are distributed? – Some many times, some only once

224571 really	38 really2	12 reaaaaaally	
1189 rly	37 reaaaaally	12 rreally	1 rreeeeeeaaaaallllllllyyyyyy
1119 realy	35 reallyyyyy	11 reaallyy	1 rreeeeeeallly
731 rlly	31 reely	11 reallllyyy	1 rreeeeeely
590 reallly	30 realllyyy	11 reeeallly	1 rrreallyyy
234 realllly	27 reaaly	11 reeeeallly	1 rrreeeaaalllyyy
216 reallyy	27 realllyy	10 reaaaly	1 rrreeeealllyyy
156 relly	26 realllyyyy	10 reallyreallyreally	1 rrreeeeaallllyy
146 realllllly	25 reallllllllly	9 r)eally	1 rrreeeeallly
132 rily	22 reaaallly	9 really-really	1 rrrlyyy
104 reallyyy	21 really-	9 reallys	1 rrrreeeally
89 reallllllly	19 reeaally	9 reeeeeeeally	1 rrrreeeeeeaaaaallllllllyyyyyy
89 reeeally	18 realllyyy	8 realky	1 rrrreeeeeeaaaaallllllllyyyyyy
84 reaaally	16 reaaaallly	8 reallyyyyyy	1 rrrrreeeeaaallllly
82 reaally	15 reaallly	8 reallyyyyyyy	1 rrrrreeeeaaalllyy
72 reeeeally	15 reallllllllly	8 reeeaally	1 rrrrrreally
65 reaaaally	15 realllllyy	7 r3ally	1 rrrrrrealy
57 reallyyyy	15 reallyreally	7 raelly	1 rrrrrreeeeeeeaaaallllllyyyyyyy
53 rilly	15 realyy	7 reaaaaaaally	1 rrrrrrrrrreally
50 reallllllly	14 realllyyyy	7 reallllllllllly	
48 reeeeeally	14 reeeeeeally	7 realllllllyyy	
41 reeally	13 reeeaaally	7 reeeeaaally	

How words are distributed in natural language? Zipf's Law in Linguistics

- Zipf's Law states that in a large corpus of natural language text, the ***frequency of a word is inversely proportional to its rank in the frequency table.***
- Mathematically, $\mathrm{f} \propto \frac{1}{r}$

Or , $\mathrm{f} = \frac{C}{r}$

- If we count up how often each word (type) of a language occurs in a large corpus, and then list the words in order of their frequency of occurrence, we can explore the relationship between the frequency of a word *f* and rank its position in the list, known as its *rank r*.

Suppose:

$$C = 10,000$$

Then:

$$f(r) = \frac{10,000}{r}$$

For rank 1:

$$f(1) = 10,000$$

For rank 2:

$$f(2) = 5,000$$

For rank 5:

$$f(5) = 2,000$$

For rank 10:

$$f(10) = 1,000$$

If the rank increases by a factor of 10, the frequency approximately decreases by a factor of 10.

Suppose the word at rank 20 occurs 500 times. What is the **approximate** frequency of the word at rank 100?

Using:

$$f(r) = \frac{C}{r}$$

At rank 20:

$$500 = \frac{C}{20}$$

Therefore:

$$C = 10,000$$

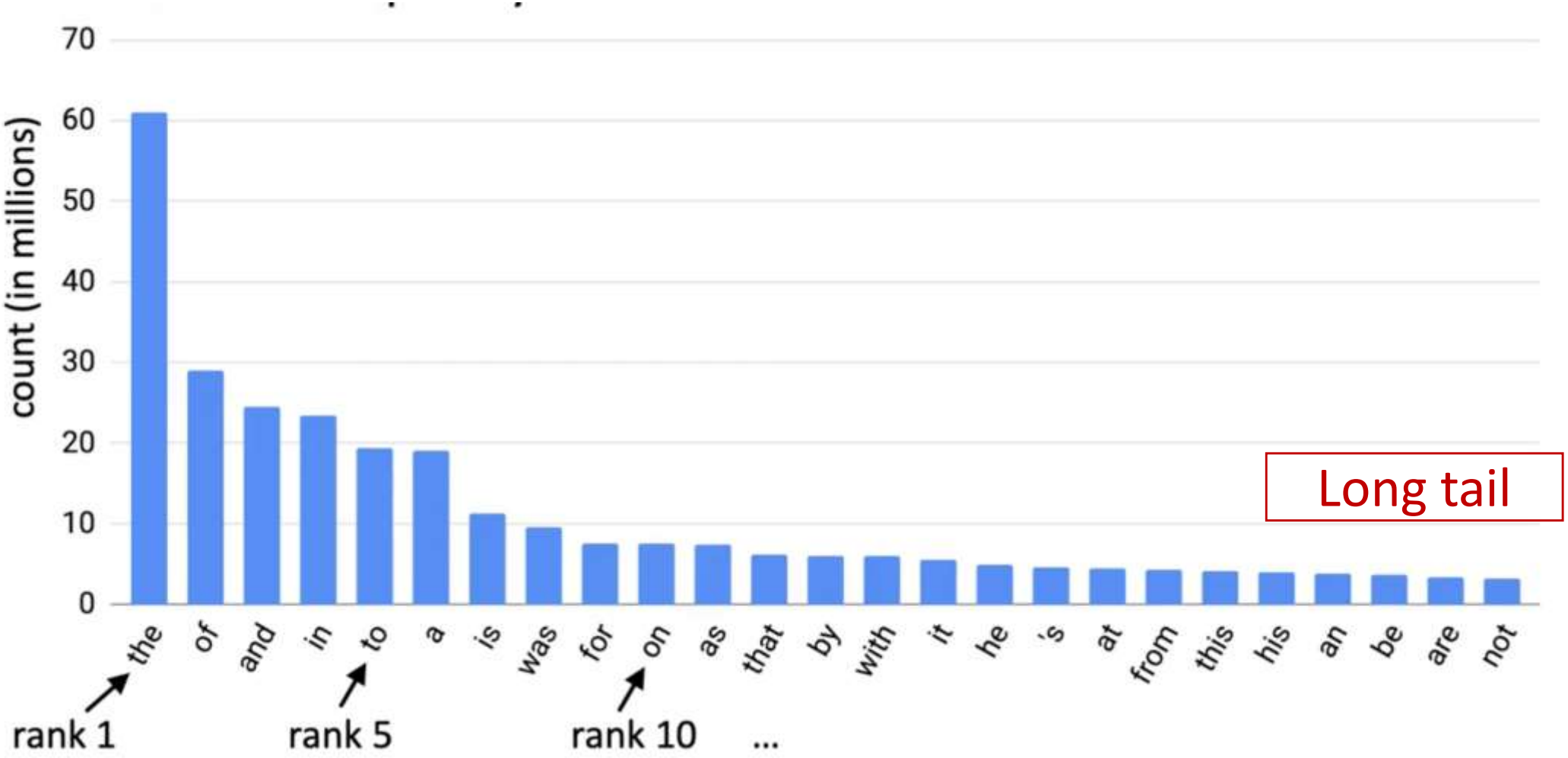
At rank 100:

$$f(100) = \frac{10,000}{100}$$

$$\boxed{100}$$

Zipf's Law is an **empirical approximation**.

Zipf's Law


70
60
50
40
30
20
10
0
count (in millions)
the
of
and
in
to
a
is
was
for
on
as
that
by
with
it
he
's
at
from
this
his
an
be
are
not
rank 1
rank 5
rank 10
...
Long tail

Long Tail Challenges in NLP

- Majority of word types are very rare so difficult to get enough data for systems to learn how to handle them
- So many word types !
- But, words have **internal structure (in addition to spelling variations), words have morphological structure like suffixes, prefixes, etc.**

224571 really	38 really2	12 reaaaaaally
1189 rly	37 reaaaaally	12 rreally
1119 realy	35 reallyyyyy	11 reaallyy
731 rlly	31 reely	11 reallllyyy
590 reallly	30 realllyyy	11 reeeeallly
234 realllly	27 reaaly	11 reeeeallly
216 reallyy	27 realllyy	10 reaaaly
156 relly	26 realllyyyy	10 reallyreallyreally
146 reallllly	25 realllllllly	9 r)eally
132 rily	22 reaaallly	9 really-really
104 reallyyy	21 really-	9 reallys
89 reallllllly	19 reeeally	9 reeeeeeeally
89 reeeally	18 realllllyyy	8 realky
84 reaaally	16 reaaaallly	8 reallyyyyyy
82 reaally	15 reaallly	8 reallyyyyyyy
72 reeeeally	15 reallllllllly	8 reeeaally
65 reaaaally	15 realllyy	7 r3ally
57 reallyyyy	15 reallyreally	7 raelly
53 rilly	15 realyy	7 reaaaaaaally
50 reallllllly	14 realllyyyy	7 reallllllllllly
48 reeeeeally	14 reeeeeeally	7 realllllyyy
41 reeally	13 reeeaaally	7 reeeeaaally

play
plays
played
playing
player
players
replay
replays
gameplay
horseplay

Zipf's Law

- A few words occur extremely frequently.
- Many of these are function words.
- Some very frequent words may contribute little topic-specific information.
- This motivates techniques such as:
  - stop-word removal
  - frequency thresholding
  - TF-IDF weighting
- But remember: Zipf's Law does not say "remove stop words".
- It only describes the frequency distribution.
- But, frequency alone is a problem!

Summary

- to do NLP on some text, we need to preprocess it:
  - tokenize documents into sentences
  - tokenize sentences into tokens
- rule-based tokenizers exist for many languages
- for writing systems without whitespace, tokenization becomes complex (often treated as an NLP problem)

Summary

- useful terms: **type**, **token**, **type/token ratio**
- when adding data, number of types keeps increasing
- most types are extremely rare (Zipf's law)

Step 1. Lexical Analysis/ Text Preprocessing

Step 1.3: Text Normalization

Process wherein different variations of text get converted to standard from

- **Spelling Correction**
- **Stemming**
- **Lemmatization**

Step 1. Lexical Analysis/ Text Preprocessing

Step 1.3: Text Normalization – Spelling Correction

- **Autocorrect – Python library**
  - **Natureal -----> Natural**

```
sentence = word_tokenize("Ntural Luanguage Processin deals with the art of
extracting insightes from Natural Languaes")
```

```
Natural Language Procession deals with the art of extracting insights from Natural Languages
```

Need spelling correctors which are aware of context

Step 1. Lexical Analysis/ Text Preprocessing

Step 1.3: Text Normalization - Stemming

Stemming - A text preprocessing technique that reduces words to their root or base form (stem) by removing affixes, simplifying text for tasks like information retrieval and text analysis

- The words "running," "runner," and "runs" would all be reduced to the stem "run"
- Running and runner ---→ runn (ing and er)
- Playing -> play; happily --→ happili
- Problem with Stemming: Stemmed words do not carry any meaning
    - e.g. : independence -> independ (which is not present in English Dictionary)
  - Lemmatization deals with such cases by using a vocabulary and analyzing the words' morphologies.
  - It returns the base forms of words which can be actually found in dictionaries

Step 1. Lexical Analysis/ Text Preprocessing

Step 1.4: Text Normalization - Lemmatization

Lemmatization - A text preprocessing technique that reduces words to their base or dictionary form (lemma) by considering the context and part of speech (POS); figuring out the most basic form or lemma of each word in the sentence.

- ‘running’ -> ‘run’
- ‘better’ -> ‘good’
- ‘went’ -> ‘go’
- ‘rocks’ -> ‘rock’
- Corpora -> corpus

- Lemmatization takes into account the grammatical category of a word (noun, verb, adjective, etc.) and provides the base form accordingly.
- For example, the lemma of “running” as a verb is “run,” while as a noun, it remains “running.

Stemming vs Lemmatization

- **Stemming is an informal process of converting words to their base forms using heuristic rules. For example, “university,” “universities,” and “university’s” might all be mapped to the base univers. (One limitation in this approach is that “universe” may also be mapped to univers, even though universe and university don’t have a close semantic relationship.)**
- **Lemmatization is a more formal way to find roots by analyzing a word’s morphology using vocabulary from a dictionary.**
- **Stemming and lemmatization are provided by libraries like spaCy and NLTK.**
- **Lemmatization considers the context and morphological analysis to return a valid word, whereas stemming applies simpler rules to chop off prefixes or suffixes, often resulting in non-dictionary words.**

Lexical Analysis

London is the capital and most populous city of England and the United Kingdom. Standing on the River Thames in the south east of the island of Great Britain, London has been a major settlement for two millennia. It was founded by the Romans, who named it Londinium.

Sentence Segmentation



1.“London is the capital and most populous city of Englar and the United Kingdom.”
2.“Standing on the River Thames in the south east of the island of Great Britain, London has been a major settlement for two millennia.”
3.“It was founded by the Romans, who named it Londinium.”

Tokenization



“London”, “is”, “ the”, “capital”, “and”, “most”, “populous”, “city”, “of”, “England”, “and”, “the”, “United”, “Kingdom”, “ .”

- https://www.deeplearning.ai/resources/natural-language-processing/